

Expressions

$$\rightarrow_e \subseteq (\text{Exp} \times \text{State}) \times (\text{Exp} \times \text{State})$$

$$(W\text{-EXP.LEFT}) \quad \frac{\langle E_1, s \rangle \rightarrow_e \langle E'_1, s' \rangle}{\langle E_1 +_s E_2, s \rangle \rightarrow_e \langle E'_1 +_s E_2, s' \rangle}$$

$$(W\text{-EXP.RIGHT}) \quad \frac{\langle E, s \rangle \rightarrow_e \langle E', s' \rangle}{\langle n +_s E, s \rangle \rightarrow_e \langle n +_s E', s' \rangle}$$

$$(W\text{-EXP.VAR}) \quad \frac{}{\langle x, s \rangle \rightarrow_e \langle n, s \rangle} \quad s(x) = n$$

$$(W\text{-EXP.ADD}) \quad \frac{}{\langle n_1 +_s n_2, s \rangle \rightarrow_e \langle n_3, s \rangle} \quad n_3 = n_1 + n_2$$